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# $SiO_2$/ITO Transparent Contacts with Anti-Soiling Coatings for Bifacial Perovskite Solar Cells in Harsh Environments

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# Abstract

Indium Tin Oxide (ITO) thin films are transparent conductive materials essential for the fabrication of semi-transparent perovskite solar cells. The silica coating is exceptionally anti-reflective, reducing sunlight reflection and anti-dust coating. This study examines the regulated modification of optical and morphological characteristics of silica thin films generated via RF magnetron sputtering. ITO films require elevated temperature processes (> 200°C) to enhance crystallinity, optical transparency, and electrical conductivity. Their thermal sensitivity restricts their application on heat-sensitive substrates. This study demonstrates that dust accumulation on silica coatings is negatively correlated with oxygen content, hence validating their self-cleaning properties. High-quality ITO layers at diminished processing temperatures are crucial for the efficiency and scalability of ST-PSCs. This study presents a room-temperature sputtering deposition of a $SiO_2$/ITO layered structure enabling transparent ST-PSC connections. Films with optimal surface morphology and thickness attain 90% visual transmittance (400–1000 nm) and sheet resistance ≤ 45.0 Ω/sq. Industrial-scale ITO coatings on 4-inch silicon substrates at ambient temperature mitigate the limitations of high-temperature methods. Films produced under low pressure (2 mTorr) and RF power (100 W) exhibited superior electrical and topological characteristics, with AFM verifying surface roughness of less than 1 nm. Hall effect measurements, X-ray diffraction, X-ray photoelectron spectroscopy, transmission electron microscopy, and atomic force microscopy validated the films' compact, homogenous, and defect-free structure. These advancements facilitate the regulated manufacture of transparent conductive oxide layers on temperature-sensitive substrates, enabling flexible and efficient optoelectronic devices.


# Introduction

Smart windows, BIPV, interactive displays, and tandem solar modules are possible with ST-SCs. They generate electricity and send light, making them vital to multifunctional energy-harvesting systems [1–3]. See-through electrodes with strong electrical conductivity and optical transparency are needed for ST-SCs. The most popular transparent conductive oxide (TCO) is indium tin oxide (ITO) because it works well with light and electronics [4–15]. Making clear rear connections is difficult, especially when deposition must stay below 50°C to safeguard essential layers. At low temperatures, ITO films form an amorphous phase, making them less transparent and conductive. These limitations have led researchers to metal nanowires, graphene, and oxide/metal/oxide (OMO) multilayer structures, which function better and can be employed at low temperatures. The front and back illumination of a MoOx/Ag/WOx stack yields 15.4% and 9.7% power conversion efficiency [16]. Clear ITO and low electrical resistance make ITO/Metal/ITO (IMI) stacks popular in organic and flexible devices. IMI structures perform well as back connections in other instances, but ST-PSCs rarely use them [16]. The study examines $SiO_2$/ITO layer stacks as transparent contacts for ST-PSCs. ITO has low resistance, excellent conductivity, and efficient optical transmission. Because of its chemical stability and high bandgap of 3.5–4.3 eV, solar cells, OLEDs, displays, and sensors use it [17–34]. Silicon dioxide ($SiO_2$) is a popular dust-proof coating for photovoltaic (PV) panels, optical equipment, and automobile parts due to its chemical stability,

hydrophobic surface, and porous nature. These coatings repel dust, and rain or wind clean them. $SiO_2$ coatings on PV systems prevent dust buildup, enhancing performance and clarity.

We fabricated 100 nm thick $SiO_2$ films using RF sputtering and modified their characteristics for anti-reflection (AR) and anti-soiling (AS) coatings. Solar applications require these coatings to prevent dust from clogging them and improve performance, especially in extreme situations. Dust storms can reduce PV module performance by 22%, and even a thin covering wastes energy. To eliminate light reflection and contamination, ARAS coatings adjust film thickness and refractive index to transmit light. Our porous $SiO_2$ coatings effectively absorb light and reduce PV system maintenance due to their AR and AS qualities. These advances enable affordable, high-performance next-generation PV technologies. We also studied how sputtering conditions affect these coatings, which was useful for industry. Figure 1 illustrates the utilization of $SiO_2$/ITO layer stacks in perovskite solar cells. Our research reveals that IAI and ICI architectures improve ST-PSC, which improves energy-harvesting devices. We improved film transparency, conductivity, and smoothness by fine-tuning deposition parameters. They were ideal for flexible, high-efficiency devices. The paper introduces a room-temperature $SiO_2$/ITO transparent electrode that improves the optical, electrical, and surface properties of semi-transparent perovskite solar cells (ST-PSCs). A $SiO_2$ layer underneath the ITO film enhances optical transmittance, reduces sheet resistance, and provides anti-reflective and anti-soiling qualities for optimal performance in real life. This dual-functionality structure was achieved by RF magnetron sputtering without post-deposition annealing. It advances high-performance transparent electrodes on temperature-sensitive substrates. As an insulator, adding a $SiO_2$ layer can alter the sheet resistance of ITO films. The study uses in-situ deposition and thin $SiO_2$ layers (~ 50 nm) to achieve flawless film interfaces (impact mitigation). Changing sputtering parameters, such as chamber pressure and RF power, improves the electrical performance of the ITO layer, compensating for resistive losses from the $SiO_2$ layer. The $SiO_2$ layer thickness impacts light transmission in the stack. Thinner $SiO_2$ coatings improve anti-reflective performance by reducing reflection and improving light coupling. Thicker $SiO_2$ layers disperse and absorb more light, leading to optical interference and hindering light transmission. Further study is needed to determine the optimal thickness of $SiO_2$ for optimal device performance, balancing anti-reflective characteristics and electrical conductivity. Flexible electronics built of heat-sensitive materials like PET can't withstand high-temperature post-sputtering treatments, making them preferable for electrical and optical applications. To generate 150 nm thick ITO films with better electrical and optical properties, we carefully altered the RF magnetron sputtering power and deposition pressure at room temperature.

# Methodology

The ITO and $SiO_2$ sputtering targets used in this study were obtained from Testbourne Ltd, with 200 mm diameter and 6 mm thickness. Transparent conductive oxides (TCOs) were deposited using RF magnetron sputtering (Oxford International PlasmaPro 400D™) with an ITO target of 99.99% purity. The indium-tin oxide (ITO) targets consisted of $In_2O_3$ and $SnO_2$ mixed in mass ratios of 90:10 (ITO90) and 50:50 (ITO50), respectively. This one-step deposition method is widely favored in research due to its

efficiency and ability to achieve high-throughput TCO fabrication, making it an ideal choice for this study. The ITO films were deposited on undoped, single-side polished 4-inch silicon wafers. The deposition base pressure was below $3 \times 10^{-7}$ Torr using a turbomolecular pump. Before depositing the ITO layer, a 50 nm silicon dioxide ($SiO_2$) layer was added using an $SiO_2$ target without breaking the vacuum. The atomic concentrations of $SiO_2$ targets used for sputtering are approximately 33.3 at.% silicon (Si) and 66.7 at.% oxygen (O), based on the stoichiometric 1:2 atomic ratio. This $SiO_2$ layer had two key roles: it passivated the substrate surface to minimize defects and improved the adhesion of the ITO film. The deposition was carried out at room temperature, with pressures ranging from 2 mTorr to 10 mTorr, maintained by adjusting the argon valve. No substrate rotation was used during deposition. The resulting ITO films, approximately 100 nm thick, showed uniform structural, electrical, and surface properties across their surface. These optimized characteristics make films suitable for various applications. Table 1 provides a summary of the deposition parameters, including RF power, deposition pressure, and deposition duration. The repeatability of the $SiO_2$/ITO thin films fabricated via RF magnetron sputtering was ensured through systematic control of key deposition parameters such as RF power, chamber pressure, and deposition time. The consistency of film properties across multiple deposition runs—evidenced by uniform surface morphology ($Ra < 1$ nm), stable optical transmittance (~ 90%), and low sheet resistance (~ 45 Ω/sq)—demonstrates the high reproducibility of the fabrication process. Furthermore, comprehensive characterization using AFM, FESEM, XRD, XPS, TEM, and electrical measurements consistently confirmed the structural and functional uniformity of the films. These findings underscore the robustness of the deposition method and enhance the overall reliability of the results, making the proposed $SiO_2$/ITO structure a promising and scalable solution for transparent conductive coatings in optoelectronic applications.

| Serial | Targets | Deposition power (watts) | Time (minutes) | Pressure (mTorr) |
|---|---|---|---|---|
| 1 | $SiO_2$ | 300 | 10 | 2 |
| | ITO | 100 | 20 | 2 |
| 2 | $SiO_2$ | 300 | 10 | 2 |
| | ITO | 200 | 13 | 2 |
| 3 | $SiO_2$ | 300 | 10 | 2 |
| | ITO | 300 | 7 | 2 |
| 4 | $SiO_2$ | 300 | 10 | 2 |
| | ITO | 400 | 5 | 2 |
| 5 | $SiO_2$ | 300 | 10 | 2 |
| | ITO | 500 | 3 | 2 |
| 6 | $SiO_2$ | 300 | 10 | 2 |
| 7 (Reference) | ITO | 100 | 20 | 2 |
| 8 (Reference) | $SiO_2$ | 300 | 10 | 10 |
| | ITO | 100 | 13 | 10 |
| 9 | $SiO_2$ | 300 | 10 | 10 |
| | ITO | 200 | 7 | 10 |
| 10 | $SiO_2$ | 300 | 10 | 10 |
| | ITO | 300 | 5 | 10 |
| 11 | $SiO_2$ | 300 | 10 | 10 |
| | ITO | 400 | 3 | 10 |
| 12 | $SiO_2$ | 300 | 10 | 10 |
| | ITO | 500 | 20 | 10 |
| 13 (Reference) | ITO | 100 | 20 | 10 |
| 14 (Reference) | $SiO_2$ | 300 | 10 | 10 |

# Thin film characterization:

The structural properties of the films were analyzed using a Rigaku™ X-ray diffraction (XRD) system. Additional structural analysis was carried out with X-ray photoelectron spectroscopy (XPS) on the Escalab 250Xi™ system (Thermo Fisher Scientific™). The XPS data was processed and fitted using Avantage software, with monochromatic Al Kα radiation (1486 eV) as the source. Narrow scans were performed at a pass energy of 20 eV, while survey scans were done at 100 eV. High-resolution spectra were collected over 10 scans, and single scans were used for survey spectrum. The optical properties of the annealed films were examined using UV-Vis-NIR spectroscopy (Jasco V100TM). Electrical properties were measured at room temperature with the Lakeshore 8404™ Hall effect system. Surface morphology was studied using a JEOL 7610™ field-emission scanning electron microscope (FESEM). For topographical analysis of the ITO films, atomic force microscopy (AFM) in Tapping mode was performed with the Bruker™ Icon Dimension system. High-resolution transmission electron microscopy (HR-TEM) was also used for film analysis. Cross-sectional TEM lamellas, with thicknesses between 40 and 70 nm, were prepared using the SEM/FIB Versa 3D dual-beam system (FEI, Hillsboro, Oregon, USA). TEM investigations were conducted on a TALOS system (FEI, Hillsboro, Oregon, USA) operating at 200 kV and equipped with an energy-dispersive X-ray (EDX) detector and a high-angle annular dark-field (HAADF) detector. The hydrophilicity of the films was evaluated through contact angle measurements using the Kruss™ system. These detailed characterizations provided valuable insights into the quality and properties of the deposited thin films. Optical measurements were conducted using a UV-Vis-NIR spectrophotometer with an accuracy of ± 1.0% in transmittance and reflectance values across the 250–1500 nm range. This ensured reliable assessment of the optical properties of the ITO and $SiO_2$/ITO films, particularly the average transmittance, which consistently reached up to 90%. For electrical performance, Hall effect measurements were performed using a van der Pauw configuration under a controlled magnetic field (± 0.5% calibration error), enabling accurate extraction of carrier concentration, mobility, and type (n-type). Sheet resistance was measured using a four-point probe system with an instrument uncertainty of ± 0.2 Ω/sq, providing high confidence in the reported values as low as 45 Ω/sq. Surface morphology was evaluated using atomic force microscopy (AFM) and field emission scanning electron microscopy (FESEM). AFM scans had a vertical resolution below 0.1 nm and lateral resolution of 10 nm, allowing precise quantification of roughness values (Ra < 1 nm). FESEM imaging provided nanometer-scale detail on film uniformity and grain structure, supporting morphological interpretations. Contact angle measurements were conducted using a calibrated goniometer with a precision of ± 1°, ensuring robust evaluation of wettability. Collectively, the accuracy and consistency of these measurements confirm the high performance and reliability of the developed $SiO_2$/ITO films, reinforcing their potential for real-world applications in photovoltaics and optoelectronics. The measurement protocols and instruments used align with industrial standards, ensuring that the reported outcomes are both scientifically rigorous and technologically relevant.

## Results and discussions

# XRD analysis:

X-ray diffraction (XRD) analysis was performed on the as-deposited ITO films over a diffraction angle range of 3° to 90°. Figure 2 shows the XRD spectra for films grown at room temperature under different conditions. The films were deposited with RF power levels of 100 W and 500 W. The analysis revealed a dominant growth along the (222) crystal plane. For films deposited at a lower pressure of 2 mTorr with reduced RF power, the intensity of the (222) peak, located around 31°, was notably higher. Additional diffraction peaks were observed at 21.5°, 31.5°, 35.5°, 51°, and 56.5°, while a strong peak near 34° was attributed to the silicon substrate. This has been reported previously [35]. At 2 mTorr deposition pressure and 500 W RF power, the intensity of the (222) peak showed significant enhancement. This improved crystallinity is likely due to grain coalescence during film growth, where smaller nuclei merge, reducing interfacial energy. Lower RF power allows atoms sufficient mobility to reorient within the lattice, promoting better alignment, minimizing structural defects, and enhancing crystallinity. The ordered crystal structure resulting from these conditions improves the films' electrical and optical properties, making them ideal for use as transparent conductive layers in optoelectronic devices. XRD patterns of ITO films show peaks corresponding to planes such as (211), (222), (400), and (440), characteristic of the cubic $In_2O_3$ bixbyite structure. The dominant diffraction peaks observed at ~ 31.5°, 35.5°, and 51° correspond well to the (222), (400), and (440) planes of the cubic $In_2O_3$ bixbyite phase, which align with ICDD PDF card No. 06-0416. The diffraction data was cross-referenced with the International Centre for Diffraction Data (ICDD) database to confirm the crystallographic phase and identify any secondary phases. This analysis provided critical insights into the structural characteristics of the films.

## XPS analysis:

Figure 3 shows the XPS analysis processed using Avantage software, confirming that the oxide films are highly pure with minimal carbon contamination, even after exposure to air. The deposition pressure had a clear impact on surface oxidation, with films deposited at 2 mTorr and 10 mTorr displaying stoichiometric oxide formation. The atomic percentages of oxygen and other elements were quantified using the method outlined by Biesenger et al. The calculated work function for both samples was approximately 3.94 eV, aligning well with values reported in previous studies. The XPS spectra were analyzed and deconvoluted to examine the core-level regions of (a) In 3d, (b) Sn 3d, (c) O 1s, and (d) C1s [35, 36]. A Voigt profile was used for fitting the data after removing the Shirley background. In the C1s spectrum, a C-N bond was observed at 285.5 eV, associated with the ITO layer, along with oxidation components (C-O and C = O) at 286–288 eV and a $\pi \rightarrow \pi^*$ satellite at 291.1 eV. The N1s spectrum displayed peaks for C-N and -NH2 bonds at 398.0 eV and 399.7 eV. The In 3d spectra revealed surface $InO_2^-$, likely formed by the diffusion and oxidation of $In^{2-}$ ions. A deeper $InO_2^-$ signal was linked to interactions between $In^{2+}$ and $O^{2-}$ ions, representing the primary structure throughout the film's thickness. These results showcase the precise chemical interactions and stoichiometric control achieved during the deposition process, contributing to the high-quality characteristics of the films.

To ensure the accuracy and reproducibility of the XPS spectral interpretation, the fitting process employed standardized methods widely accepted in surface chemistry analysis. Specifically, the Shirley

background subtraction was applied to account for inelastic background signals arising from photoelectron scattering, offering a consistent baseline across all core-level spectra. Additionally, Voigt profiles—combinations of Gaussian (instrumental broadening) and Lorentzian (natural line width) components—were used to fit the spectral peaks. This hybrid function is particularly effective in capturing the true peak shape and minimizing fitting errors. Peak deconvolution was performed using fixed or physically justified FWHM ranges, and binding energy positions were cross validated with standard literature values to ensure consistency. These fitting strategies, along with repeated scans and software-based fitting residual checks, contribute significantly to the reliability and scientific rigor of the XPS data presented in this study.

The analysis includes the silicon (Si 2p) and oxygen (O 1s) spectra collected after monatomic etching, along with oxygen profiling results for the thin films. To examine the chemical states of oxygen, carbon spectra fitting was used to identify oxygen associated with carbon species, such as C–O and C = O bonds. This method helped isolate oxide-related oxygen, representing metal-oxygen bonds, from the total oxygen content. The spectra primarily showed signals for silicon and oxygen, with distinct SiOx chemical states. The Si 2p peak was observed at 103 eV. The stoichiometric composition of the SiOx films was determined to be $SiO_{1.91}$. The study also demonstrated that the silica stoichiometry remained consistent across different etching techniques, including cluster and monatomic etching. These findings underline the precise control over oxygen incorporation and the resulting silicon chemical states in the films.

## Optical characterization:

The optical properties of $SiO_2$/ITO thin films developed via RF magnetron sputtering were systematically investigated using UV-Vis-NIR spectroscopy, as illustrated in Figs. 4 and 5. The films were deposited under two distinct working pressures (2 mTorr and 10 mTorr) and across a range of RF powers (100 W to 500 W) to assess their transmission and absorption behavior. The spectrophotometric results revealed clear dependencies of optical performance on the deposition parameters. Films fabricated at the lower pressure of 2 mTorr consistently exhibited higher transparency throughout the visible spectrum, achieving average transmittance values above 85%. This enhancement is attributed to the formation of a denser and more homogeneous film microstructure, which reduces light scattering and minimizes optical losses. In contrast, films sputtered at 10 mTorr showed a moderate decrease in transmittance, likely due to increased surface roughness and defect formation caused by more frequent collisions in the higher-pressure plasma environment. Similarly, RF power had a significant influence on the films' optical behavior. Lower power levels, particularly at 100 W, resulted in films with greater transparency and reduced absorption. This can be linked to slower deposition rates and improved atomic arrangement, which help minimize structural defects and impurities. Conversely, higher RF power (500 W) led to slightly diminished optical performance, possibly due to the increased kinetic energy of sputtered species, which can introduce imperfections in the growing film. The calculated absorption values—derived from the difference between incident and transmitted light—confirmed minimal

absorption in the visible range for all samples. This low absorption, combined with high transmittance, underscores the potential of these $SiO_2$/ITO films for applications requiring excellent optical clarity, such as transparent electrodes in photovoltaic modules, displays, or optoelectronic devices. Overall, these findings highlight the critical role of sputtering conditions in tailoring the optical performance of $SiO_2$/ITO thin films. Careful optimization of deposition pressure and power is essential to achieving the desired balance between transparency and structural integrity, enabling their effective integration into next-generation energy and electronic technologies. In the context of the UV-Vis-NIR measurements, thinner films typically exhibit higher optical transmittance due to reduced absorption and scattering, which aligns with our observation of transmittance values exceeding 85% for films deposited under lower RF power and shorter deposition times. As film thickness increases, absorption generally rises in the near-infrared region due to the extended path length and increased carrier density, leading to a gradual decline in transparency. This trend is complemented by Hall effect measurements, where increased thickness often enhances electrical conductivity by reducing surface and interface scattering effects, thus facilitating better carrier mobility and continuity across grain boundaries. However, excessively thick films may introduce structural defects or stress that hinder mobility and increase resistivity. Therefore, an optimal balance is required—where moderate thickness supports adequate free carrier transport for low sheet resistance, while still maintaining high transmittance in the visible spectrum. Clarifying this thickness-property relationship supports the design of transparent electrodes that meet both optical and electronic requirements for photovoltaic applications.

# Contact angle characterization:

Figure 7 presents the contact angle measurements of $SiO_2$/ITO thin films fabricated via RF magnetron sputtering, offering insights into their surface wettability and how it is influenced by sputtering conditions. The measurements were conducted on films deposited at two working pressures (2 mTorr and 10 mTorr) and across five RF power levels (ranging from 100 W to 500 W). A clear correlation was observed between contact angle behavior and both deposition pressure and RF power. Films grown at the lower pressure of 2 mTorr consistently exhibited smaller contact angles, indicating enhanced hydrophilicity. This behavior can be attributed to smoother, denser films with fewer surface defects—conditions promoted by the reduced gas-phase scattering at low pressure, which facilitates compact and uniform film growth. Conversely, at 10 mTorr, the contact angles increased, suggesting a shift toward more hydrophobic surfaces. This is likely due to increased roughness and impurity incorporation arising from more frequent collisions in the plasma, which reduce the film's surface energy and hinder wetting. Similarly, RF power had a notable impact on wettability. Films deposited at 100 W displayed the most hydrophilic behavior, likely due to slower deposition rates and more orderly surface formation. In contrast, films grown at higher RF power (up to 500 W) showed larger contact angles, indicating rougher and less uniform surfaces. These findings underscore that both sputtering pressure and RF power must be carefully optimized to engineer the desired surface energy of $SiO_2$/ITO coatings, especially in applications where anti-soiling or self-cleaning properties are critical for long-term device performance.

# FESEM characterization:

Field Emmision Scanning Electron Microscopy (FESEM) was employed to investigate the surface morphology and microstructural properties of $SiO_2$/ITO films fabricated via RF magnetron sputtering under two different pressures (2 mTorr and 10 mTorr) and five RF power levels (100 W to 500 W) as shown in Fig. 7. FESEM analysis revealed that deposition conditions significantly affected the grain size, surface uniformity, and overall morphology of the films. At the lower pressure of 2 mTorr, the films exhibited a denser and more compact surface with uniform grains. This can be attributed to the reduced scattering of sputtered particles, allowing for more precise deposition on the substrate. At higher pressures of 10 mTorr, the films showed increased surface roughness and larger grain boundaries, indicating a less uniform deposition process. This is likely due to enhanced gas-phase collisions at higher pressures, which reduce the energy of the arriving particles and lead to less compact film growth. Additionally, the RF power had a noticeable impact on the morphology. These observations from FESEM analysis provide critical insights into the correlation between sputtering conditions and the microstructural evolution of $SiO_2$/ITO films.

# AFM characterization:

Atomic Force Microscopy (AFM) was used to analyze the surface topography and roughness of $SiO_2$/ITO films prepared via RF magnetron sputtering under two deposition pressures (2 mTorr and 10 mTorr) as shown in Fig. 8 and five RF power levels (100 W to 500 W). The AFM measurements revealed distinct variations in surface roughness and morphological characteristics depending on the sputtering conditions. Films deposited at the lower pressure of 2 mTorr exhibited smoother surfaces with a root mean square (RMS) roughness typically below 1 nm. This smoothness is attributed to the higher energy of sputtered particles, which results in better surface diffusion and uniform film growth. At the higher deposition pressure of 10 mTorr, the films demonstrated increased RMS roughness, indicating a less homogeneous surface topology. This change is likely due to enhanced gas-phase scattering, reducing the kinetic energy of the particles and leading to irregular film growth. The influence of RF power was also significant. At lower RF power levels (200 W), the films exhibited finer and more uniform surface features, consistent with a slower deposition rate that allows for controlled nucleation and grain growth. As the RF power increased to 800 W, the surface morphology became rougher, with a noticeable increase in grain size and peak-to-valley distances. This trend suggests that higher power increases the kinetic energy of the sputtered particles, promoting rapid growth but at the expense of surface smoothness. These AFM observations underscore the critical role of sputtering parameters in tailoring the surface properties of $SiO_2$/ITO films. The results provide valuable insights for optimizing film characteristics to meet the requirements of specific applications, such as in optoelectronics and transparent conductive coatings.

# TEM characterization:

Transmission Electron Microscopy (TEM) analysis of $SiO_2$/ITO films deposited using the sputtering technique under two pressures (2 mTorr and 10 mTorr) and five RF power levels (100 W, 200 W, 300 W, 400 W, and 500 W), provided an in-depth evaluation of the films' structural and morphological characteristics as shown in Figs. 9 and 10. TEM imaging revealed a uniform and compact film structure with well-defined grain boundaries, confirming high-quality deposition. At lower deposition pressure (2 mTorr), the films exhibited larger, well-aligned grains with fewer crystalline defects, especially at lower RF power levels (200 W). These conditions facilitated the growth of more ordered structures due to lower ion bombardment energy, reducing interstitial defects and dislocations. As the RF power increased, the TEM analysis revealed a denser film with smaller grains, attributed to higher energy particles during sputtering causing more nucleation sites but less grain growth. At higher deposition pressure (10 mTorr), the film quality was affected by increased gas scattering, leading to reduced grain size and more defects, evident from diffuse and less-defined lattice fringes in the high-resolution TEM images. The interface between the $SiO_2$ and ITO layers remained smooth and well-adhered, showcasing the efficacy of the sputtering process for achieving strong adhesion and consistent layering. Electron diffraction patterns confirmed the polycrystalline nature of the ITO films, with dominant diffraction rings corresponding to the (222), (400), and (440) planes of cubic bixbyite $In_2O_3$. The presence of fewer defects at lower RF power and lower pressure contributed to enhanced optical and electrical properties. These results underscore the critical role of deposition parameters in influencing film microstructure, paving the way for optimizing $SiO_2$/ITO films for applications in optoelectronics and photovoltaics.

# Electrical characterization:

The electrical properties of the $SiO_2$/ITO thin films were characterized using Hall effect and four-point probe measurements [35]. Hall effect analysis provided detailed insights into carrier concentration, mobility, and conductivity type, while the four-point probe method was used to assess sheet resistivity. All the films demonstrated n-type conductivity, with an average carrier concentration of approximately $5.30 \times 10^{20}$ $cm^{-3}$. The highest carrier mobility, 35.51 $cm^2/V{\cdot}s$, was recorded at the lowest deposition pressure (2 mTorr) and a moderate RF power of 200 W. Notably, the lowest resistivity, $2.79 \times 10^{-4}$ $\Omega{\cdot}cm$, was achieved at the same pressure but with a higher RF power of 500 W. Sheet resistance values varied between 40 and 90 Ω/square, increasing with rising RF power. The electrical performance of the films was strongly influenced by the sputtering conditions. An increase in RF power led to more energetic ion bombardment of the target, enhancing the deposition rate and promoting the formation of structural defects, particularly oxygen vacancies. These vacancies act as donor states, increasing the free carrier concentration and thereby improving conductivity. However, the introduction of such defects at high RF power levels also resulted in increased grain boundary scattering, reducing carrier mobility. In contrast, films deposited at lower RF powers showed lower defect densities and better structural order, which favored higher mobility despite a reduction in carrier concentration. Deposition pressure also played a critical role in determining electrical behavior. Higher pressures limit the energy of sputtered species, impairing film crystallinity and increasing the likelihood of scattering phenomena, ultimately raising resistivity. The balance between oxygen vacancy concentration and film quality is essential for optimizing electrical performance, as excessive vacancies—while enhancing conductivity—can

detrimentally impact optical transparency. These findings underscore the necessity of fine-tuning deposition parameters to control defect formation and microstructural characteristics. Such optimization is vital for tailoring the electrical properties of ITO thin films for advanced optoelectronic and photovoltaic applications, where both high conductivity and transparency are required.

# Conclusions

This study provides significant insights into the development of low-temperature, transparent conductive oxide (TCO) layers for next-generation optoelectronic devices, particularly semi-transparent perovskite solar cells (ST-PSCs). While Indium Tin Oxide (ITO) is widely recognized for its superior optoelectronic properties, conventional deposition methods demand high-temperature processing (> 200°C) to enhance crystallinity and conductivity—conditions incompatible with flexible or heat-sensitive substrates. Our findings demonstrate that by employing RF magnetron sputtering at room temperature and systematically tuning deposition parameters such as RF power and chamber pressure, it is possible to achieve highly conductive and transparent ITO films without the need for post-deposition thermal annealing. The incorporation of a silica ($SiO_2$) layer not only enhances the anti-reflective performance of the transparent electrode but also contributes to surface engineering that promotes self-cleaning behavior. These effects arise due to the tailored nanostructure and surface chemistry of $SiO_2$, which reduce van der Waals, electrostatic, and capillary interactions—known contributors to dust adhesion—thereby improving long-term optical clarity and reducing maintenance needs. Inspired by cicada wing textures, the $SiO_2$ layer's engineered surface roughness further minimizes light reflection and enables passive removal of contaminants. ITO films deposited at lower pressures (2 mTorr) and moderate RF power (100–200 W) exhibited enhanced optical transmittance (up to 90%) and lower sheet resistance (as low as 45 Ω/sq), attributed to denser microstructures and reduced scattering losses. The Hall effect measurement revealed that moderate ion energy at these conditions facilitates defect minimization, preserving carrier mobility while promoting optimal carrier concentration via controlled oxygen vacancy formation. The reduced roughness (Ra < 1 nm) measured via AFM further supports efficient charge transport and optical uniformity, crucial for photovoltaic integration. This work advances current knowledge by demonstrating that low-temperature PVD processes, when precisely tuned, can yield industrial-grade $SiO_2$/ITO structures with multifunctional performance—high transparency, excellent conductivity, anti-reflectivity, and anti-soiling properties. This study offers a comprehensive and impactful contribution to the field of optoelectronic materials by demonstrating a scalable, low-temperature method for fabricating high-performance transparent conductive oxide (TCO) films, specifically suited for semi-transparent perovskite solar cells (ST-PSCs) and other next-generation photovoltaic technologies. By employing room-temperature RF magnetron sputtering and optimizing key deposition parameters, we successfully produced ITO films with excellent optical transparency and electrical conductivity—achieving performance levels comparable to those attained via conventional high-temperature methods.

# Declarations

**CRediT authorship contribution statement:**

M. I. Hossain: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization.

Atef Zekri: Investigation, Formal analysis.

Puvaneswaran Chelvanathan: Investigation, Formal analysis.

Brahim Aissa: Supervision.

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# Figures

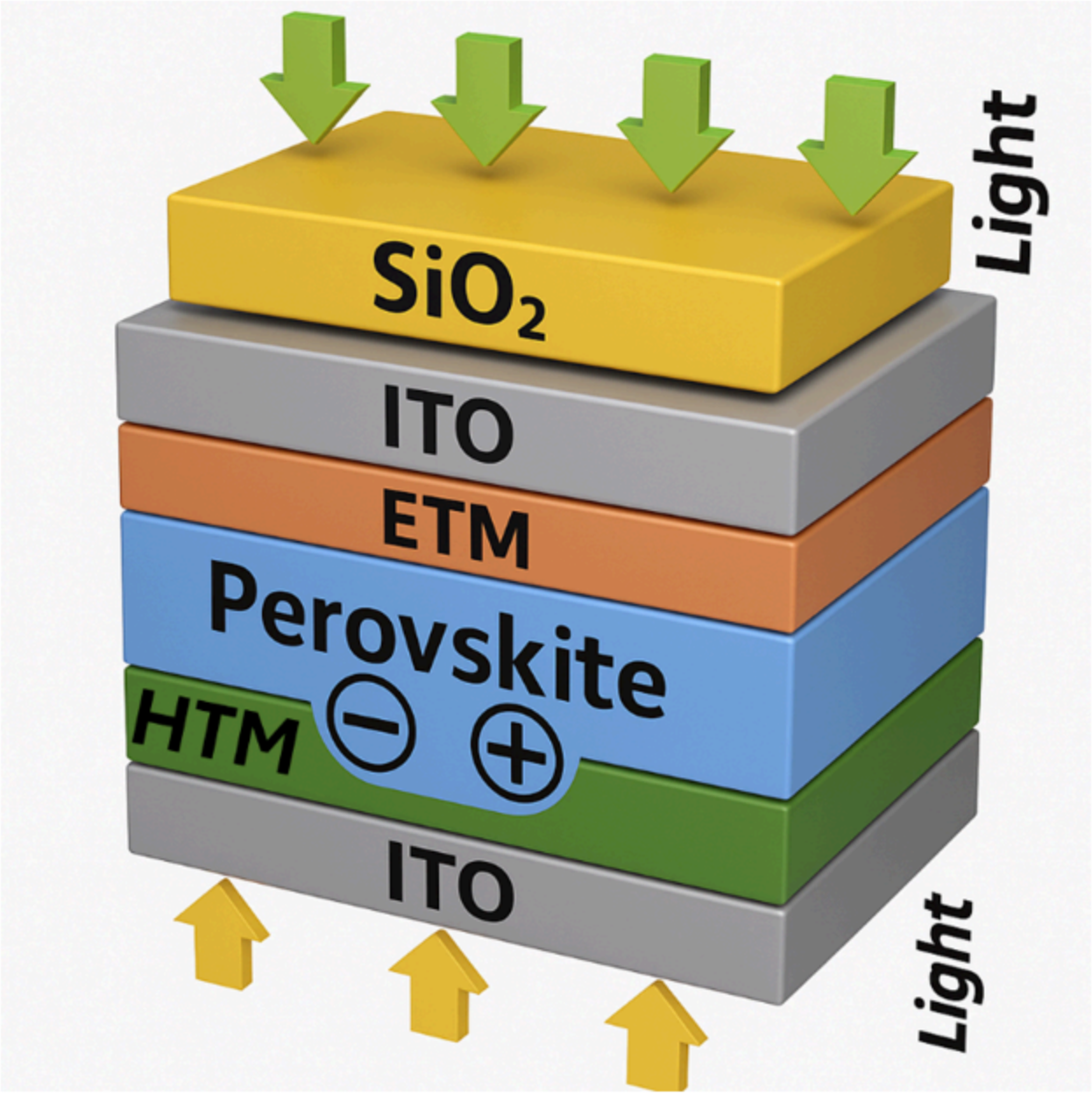
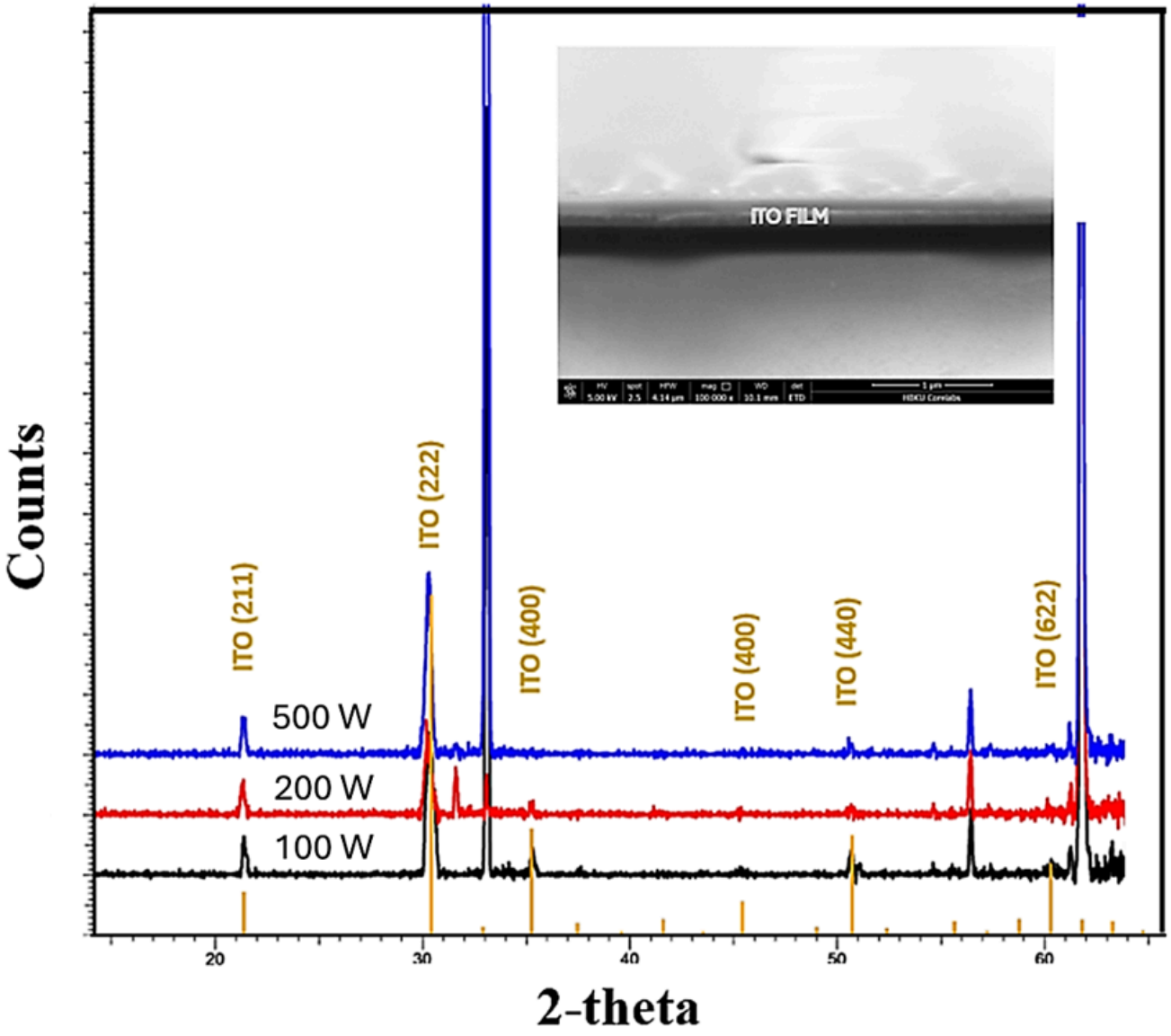



**Figure 1**

$SiO_2$/ITO structure for perovskite solar cells




## Figure 2

XRD analysis of the TCO films.

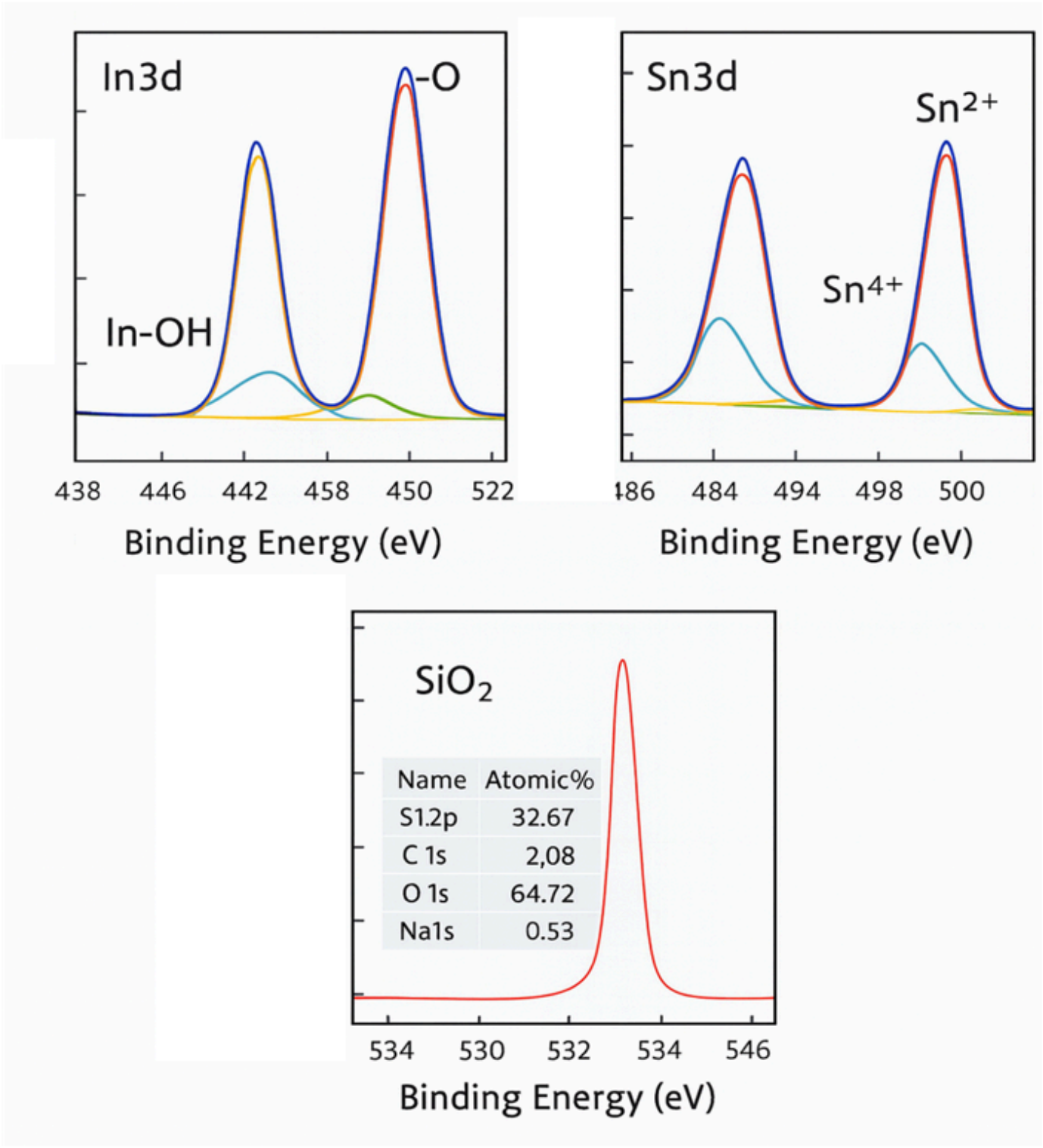
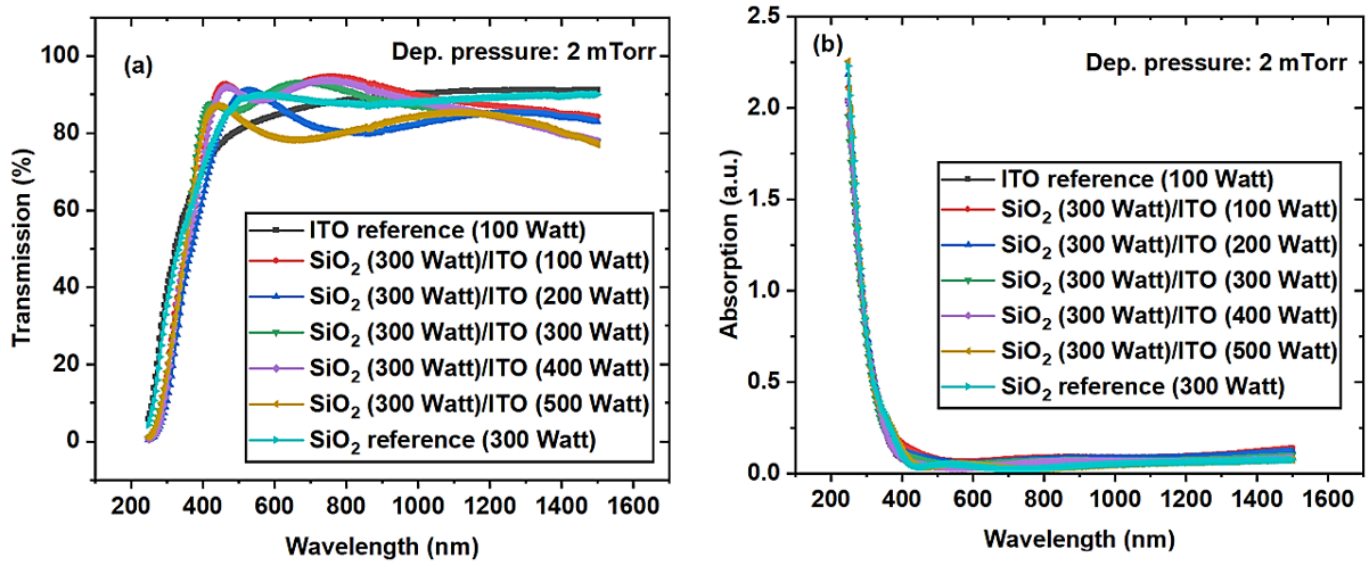


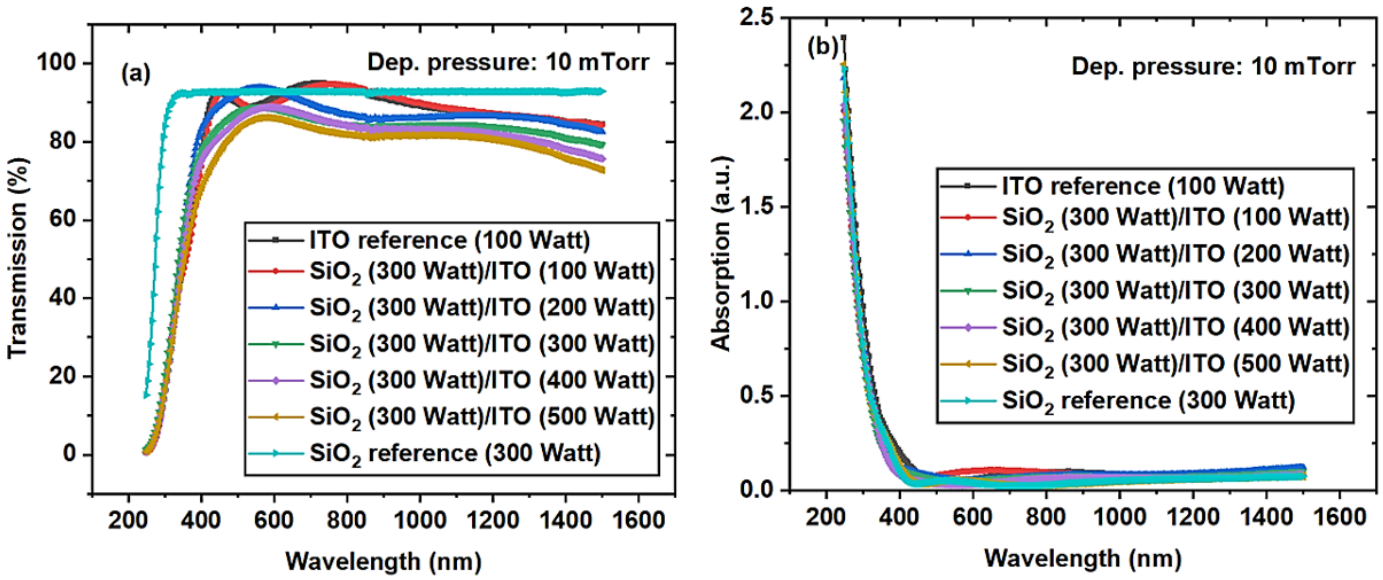

**Figure 3**

XPS spectra of ITO and $SiO_2$.




## Figure 4

Optical analysis of the stacked films.




## Figure 5

Optical analysis of the stacked films.

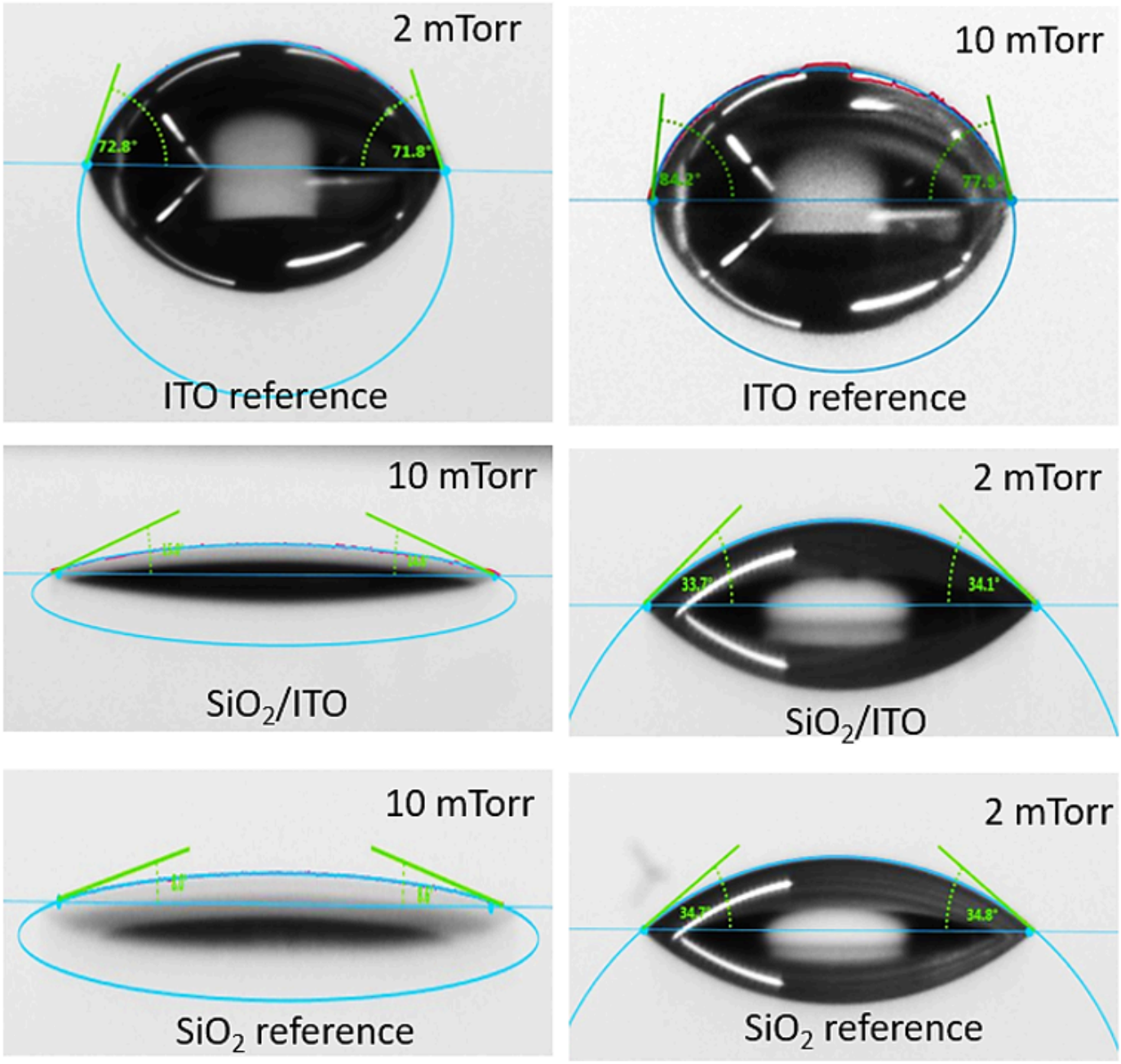
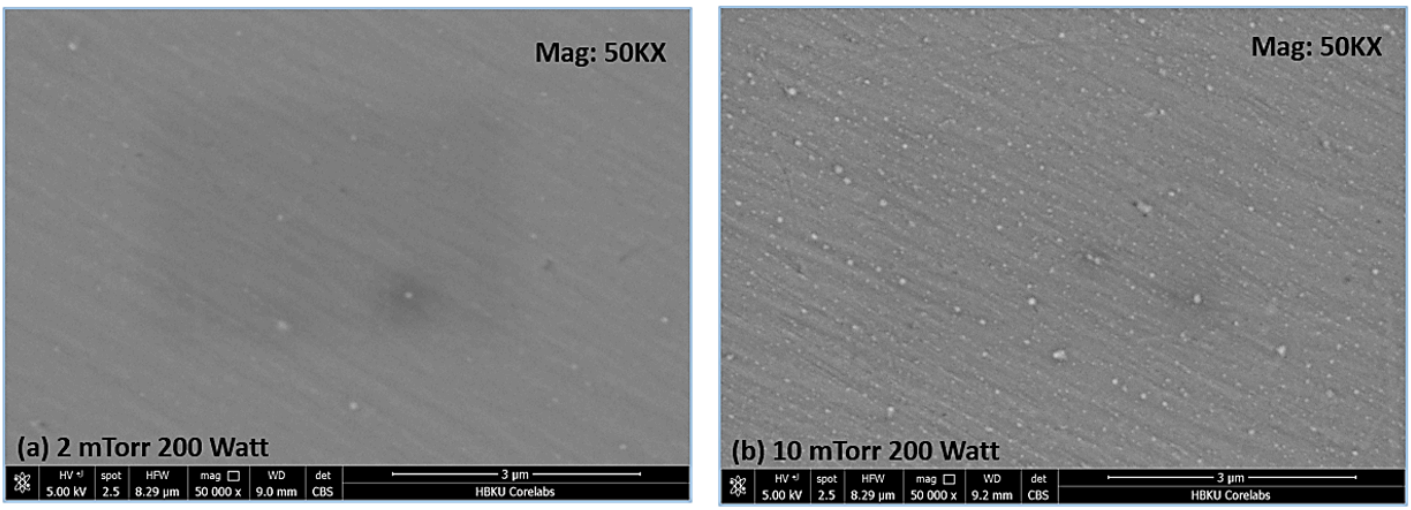


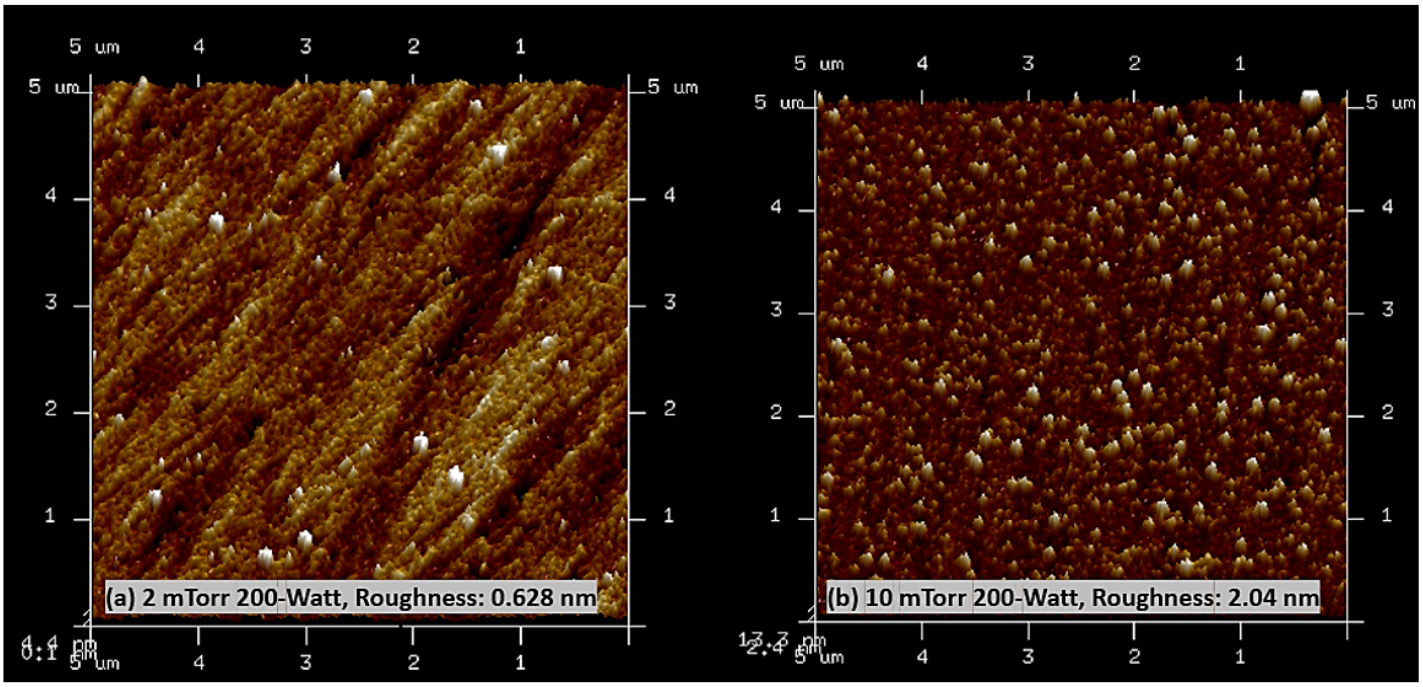

**Figure 6**

Contact angle analysis of the films.



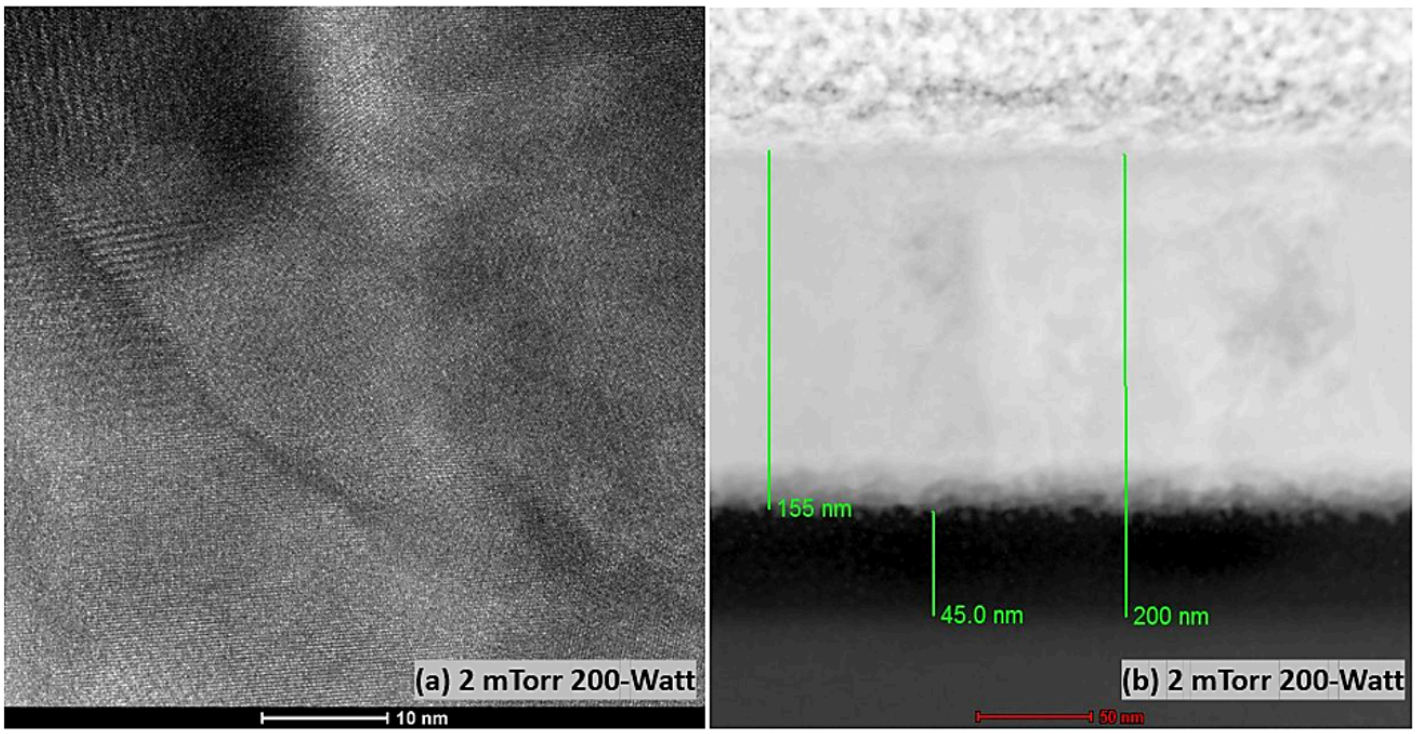

## Figure 7

SEM analysis of the films grown at two deposition pressures.



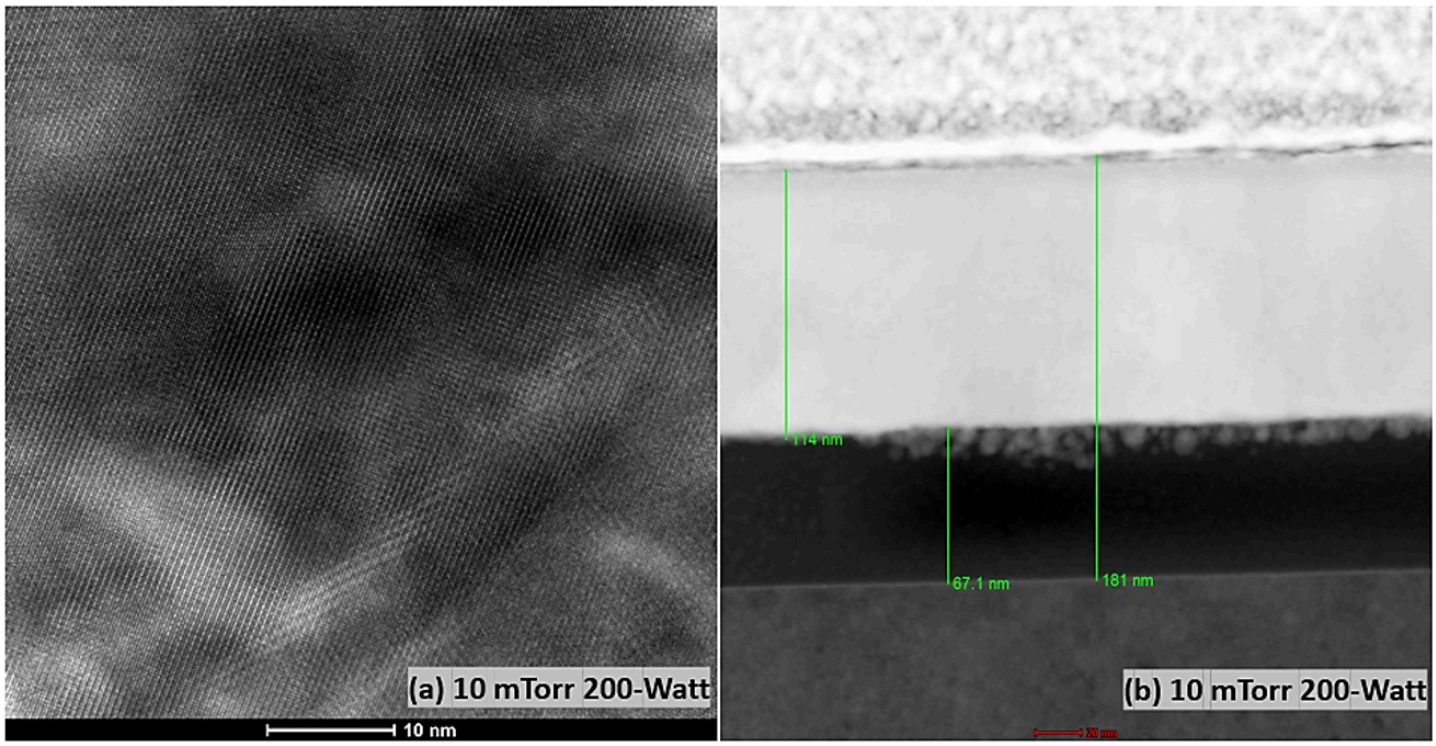

## Figure 8

AFM analysis of the films grown at two deposition pressures.




## Figure 9

TEM analysis of the films.




## Figure 10

TEM analysis of the films.